Schema and integrity assessment

Scope and current state

The complete Prisma model set and all local migration files were parsed: **211 models, 87 migrations**. The configured database resolved to a local PostgreSQL target. Catalog and aggregate queries ran in explicit READ ONLY transactions, with rollback, statement/lock timeouts and no business identifiers in results. This is local deployed-schema evidence, not production evidence.

Every model has an entry in schema-model-inventory.csv and model-ownership-register.csv. The former records fields, decimals, JSON, candidate writes/reads, lifecycle and cascades; the latter assigns a candidate purpose/domain, tenant boundary and explicit authority/retention confidence. verification/schema-model-details.json preserves the field-level inventory and source lines. Regex candidates exclude dynamic calls/raw SQL and are **not proof of exclusive authority**. A reviewed model-specific retention schedule was not established; the register does not invent one.

Intended versus local database

- All 211 model tables and scalar/enum/array column names have local catalog counterparts. No missing or extra model columns were detected.
- All 87 migration names and SHA-256 checksums match recorded applied migrations; no unfinished or pending migration was found.
- Five additional tables are accounted for: _UserRoles, _prisma_migrations, legacy_production_batches, legacy_recipe_ingredients, legacy_recipes. Legacy manufacturing tables are not evidence of an active BOM/production module.
- The catalog contains 216 tables, 55 noninternal triggers and 147 CHECK constraints.
- This comparison establishes table/column-name and migration-history parity. It does **not** prove exhaustive type/default/index/privilege parity, fresh replay, safe upgrades or production drift absence. Catalog definitions were retained for those next checks.

Evidence: verification/schema-database-comparison.json; verification/database-metadata.json; verification/migration-inventory.json. Prisma validation passed. No migrations or seed scripts ran during this audit.

Authority, identity and retention

Most business models carry organizationId. The 28 without it include authentication/global catalogs and child rows such as InventoryLevel, order lines, cash drawers and agent steps. Parent scoping can be correct; absence of organizationId is not a vulnerability by itself. Newer purchase-correction, fiscal and payroll evidence uses stronger composite tenant links and immutable lifecycle controls. Older journals, AP and several child relations still depend more heavily on service validation.

services/purchasing/ap-control.service.ts:1 checks supplier/PO scope and locks economic owners. services/hris/org.service.ts:26 resolves employee organization and effective manager/delegation scope; its legacy location-manager fallback is current-only. Financial audit data and payroll/private HR data have different access needs; model-specific purge/legal-hold policies and actual scheduler enforcement remain unverified.

Cascading organization ownership must not become an accidental financial-record eraser. F10/F14 explain the bounded risks without claiming a live exploit.

Identifiers in JSON/sourceType/sourceId fields are flexible but do not gain relational integrity automatically. The inventory lists unlinked ID-like fields and JSON fields. Their business validity must be demonstrated by source-owner commands, schema versions, hashes and migration/repair tooling. In particular, SupplierInvoice.ledgerPostingBatchId is a nullable string: link coverage must distinguish legitimate drafts from incorrectly unposted final documents.

Money, quantities and time

The parsed schema uses Decimal for money/quantities and no Float fields. Common quantities use Decimal(12,3), unit costs Decimal(12,2), totals Decimal(14,2). This avoids binary floating-point storage but still requires a reviewed rounding policy at each calculation boundary. Source amounts, tax, base amounts, rate direction and currency version must be separate immutable facts: F01/F07 show that field availability alone does not preserve meaning.

InventoryTransaction carries signed quantity, balanceAfter, effectiveAt, recordedAt and timeProvenance. These support business-time and recorded-time analysis. Legacy-createdAt approximation is explicitly represented. Backdated valuation and cutoff behavior still need isolated end-to-end tests. Partial fiscal-month clipping is defective (F08). Analytics order-activity derives organization timezone/currency and half-open windows; do not generalize that correctness to every report.

Transactions, concurrency and derived data

The inventory stock-event kernel uses versioned conditional updates and available-quantity checks; AP commands lock supplier/PO rows and record durable posting/recovery evidence. Payroll transitions use expected versions, serializable transactions and immutable snapshots/events. Period posting obtains a shared row lock, while close obtains an exclusive lock. Outboxes use bounded claims, retry counts, stale leases and dead-letter states. These are positive design facts, but mocks cannot establish PostgreSQL race behavior.

Offline queue operations lack an equivalent transactional durability boundary (F03/F04). Generic accounting reversal omits some source ownership (F02). A period lock and a current close-assurance readiness decision are not interchangeable (F06).

Read-only consistency results and denominators

Check	Observed local result	Limit
Journal balance	0 unbalanced POSTED/REVERSED entry-currency groups	Balanced does not mean economically complete or correct in base currency.
Journal tenant joins	0 line/entry/account tenant mismatches	No adversarial inserts or all-relation tenant proof.
FX storage	12 non-unit-rate lines, 0 candidate conversion mismatches	Current service reproduction still fails F01; sample may be seeded.
Journal source linkage	4 posted/reversed entries lack postingBatchId	Local observation requiring fixture/source classification, not a production-corruption claim.

Check	Observed local result	Limit
Stock projection	541 levels; 0 available/on-hand/reserved mismatches; 0 negatives; 0 approximate value mismatches	Does not prove all valuation policy scenarios.
Stock movements	541 levels all have movement evidence; signed physical movement sums agree with quantities	Opening-seed lineage and inventory-to-GL valuation completeness still need controlled proof.
AP	24 allocations with consistent tenant/supplier/currency; no overallocated draft payment	All 24 invoices and 24 payments are DRAFT; zero posted AP lifecycle denominator.
Payroll	24 run arithmetic checks agree	All 24 runs are DRAFT (with payslips); zero emitted/posted denominator. Earlier zero emitted mismatch count is therefore non-probative.
Reconciliation	24 runs	All DRAFT; no certificate hashes or sign-off.
Periods	0 overlapping pairs or out-of-year periods	Does not cover F08 creation edge case.

The sampled ledger contains opening balances, POS sale/payment and stock adjustment batches. It cannot prove purchase/payroll/reconciliation source-to-ledger completeness. Query text and aggregate outputs: verification/database-tieouts.json, verification/database-cross-domain-tieouts.json.

Performance, migration and recovery limits

Tenant/time/status indexes are present across the major fact tables, and the order-activity report uses full database aggregation rather than a truncated application slice. No EXPLAIN ANALYZE, large-tenant load test, query-count budget or production latency/cost study was run. Therefore no scalability or cost readiness is certified.

Historical local restore evidence covers 80 migrations on 29 August, not the current 87-migration runtime. Current fresh replay, upgrade from a populated prior release, trigger negative tests, recovery objectives and actual restore/login/accounting checks remain explicit release work. No rollback script should be inferred from Prisma migration presence.

Existing retention metadata

HRIS document evidence already records retentionPolicyCode, retentionBasis, retainUntil and legalHold and projects RETAIN / LEGAL_HOLD / EXPIRED_REVIEW_REQUIRED / UNDECIDED. This is a useful control and was not treated as absent. What remains unverified is a qualified, complete model-by-model retention schedule and deployed enforcement. Test fixture labels such as CM_EMPLOYMENT_10Y are not independent proof of a legally correct duration.